

FINEXA AlphaPack™ (Grey Wolf Optimizer) — User-Friendly Guide & Technical Specification

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1. What is this Module? (In Simple Words)

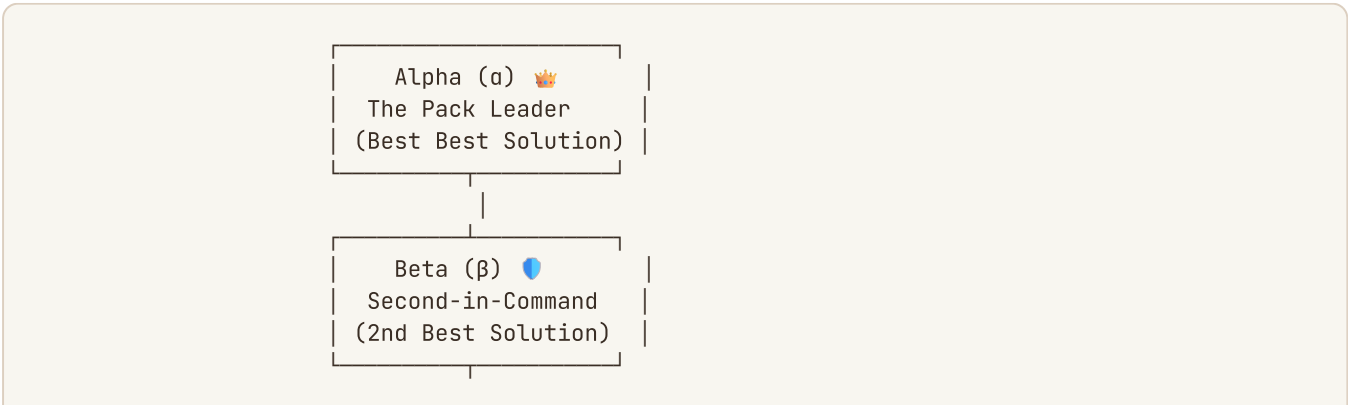
FINEXA AlphaPack™ is an AI-powered smart investment optimizer that finds the absolute best place to invest your monthly savings.

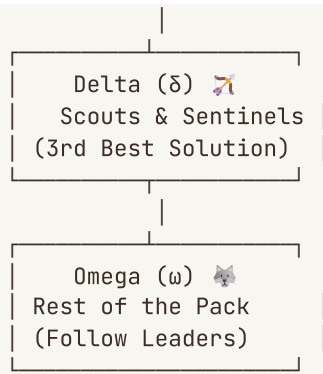
When you want to invest (for a car, home down payment, child education, or retirement), you have many choices: * Large Cap Stocks (Nifty 50) * Mid & Small Cap High-Growth Stocks * Safe Corporate Bonds & Debt * Sovereign Gold / Gold ETFs * Global / US Tech Index

Instead of guessing how much percentage to put in each, **FINEXA releases a virtual pack of 25 AI wolves**. These wolves hunt through thousands of market combinations to find the exact combination that gives you the **highest return with the lowest risk**.

2. The Nature Concept: How Grey Wolves Hunt

In nature, grey wolves live in organized packs with strict social hierarchy:





1. **Alpha (α):** The dominant leader. In FINEXA, this is the **Champion Portfolio** that achieves the highest financial score.
2. **Beta (β):** The trusted advisor. In FINEXA, this is the **Defensive Shield Portfolio** (lower risk, steady safety).
3. **Delta (δ):** The scouts. In FINEXA, this is the **Aggressive Growth Portfolio** (captures maximum upside).
4. **Omega (ω):** All other wolves follow the direction of α, β, δ towards the prey (the global optimal investment balance).

3. The 3 Pack Leaders: Alpha, Beta, Delta

When you run FINEXA AlphaPack™, it instantly presents the top 3 leader strategies:

LEADER	TITLE	ROLE & STYLE	WHO IT'S BEST FOR
Alpha (α)	The Champion	Global Best Balance: Maximizes the Sharpe ratio (highest return per unit of risk).	Everyday investors seeking balanced, steady wealth compounding.
Beta (β)	The Shield	Capital Preservation: Heavy bonds and gold buffer with lower volatility.	Short-term goals (1–3 yrs) or low risk comfort.
Delta (δ)	The Hunter	Aggressive Alpha: Heavy high-growth mid-caps and global tech equities.	Long-term goals (10–15+ yrs) targeting maximum financial freedom.

4. How the GWO Algorithm Works in Finance

4.1 Mathematical Equations of the Pack (Mirjalili et al., 2014)

During the hunt, wolves encircle the prey according to two vectors:

$$\vec{D} = |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)|$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D}$$

Where: * $\vec{X}(t)$ is the current position vector (portfolio weights across the 5 asset classes). * $\vec{X}_p(t)$ is the position of the prey (the optimal portfolio). * $\vec{A} = 2\vec{a} \cdot \vec{r}_1$

- $\vec{a}$ and $\vec{C} = 2\vec{r}_2$ are coefficient vectors. * Parameter α linearly decreases from 2.0 to 0 over iterations to balance **Exploration (searching for prey)** and **Exploitation (attacking prey)**.

4.2 Updating Position via the 3 Leaders (α , β , δ)

For every Omega wolf i , its new position is guided by the 3 leaders:

$$\begin{aligned} \vec{D}_i^{\alpha} &= \left| \vec{C}_1 \cdot \vec{X}_i^{\alpha} - \vec{X}_i \right|, \quad \vec{X}_i^{\alpha} = \vec{X}_i - \vec{A}_1 \cdot \vec{D}_i^{\alpha} \\ \vec{D}_i^{\beta} &= \left| \vec{C}_2 \cdot \vec{X}_i^{\beta} - \vec{X}_i \right|, \quad \vec{X}_i^{\beta} = \vec{X}_i - \vec{A}_2 \cdot \vec{D}_i^{\beta} \\ \vec{D}_i^{\delta} &= \left| \vec{C}_3 \cdot \vec{X}_i^{\delta} - \vec{X}_i \right|, \quad \vec{X}_i^{\delta} = \vec{X}_i - \vec{A}_3 \cdot \vec{D}_i^{\delta} \\ \vec{X}_i(t+1) &= \frac{\vec{X}_i^{\alpha} + \vec{X}_i^{\beta} + \vec{X}_i^{\delta}}{3} \end{aligned}$$

4.3 The Fitness Function (Sharpe Ratio & Return)

The fitness score of each portfolio position $\vec{w} = [w_1, \dots, w_5]$ is evaluated by:

$$\begin{aligned} \text{Expected Return } E[R_p] &= \sum_{i=1}^5 w_i \cdot \mu_i \\ \text{Portfolio Variance } \sigma_p^2 &= \vec{w}^T \cdot \mathbf{\Sigma} \cdot \vec{w} \\ \text{Sharpe Ratio } S_p &= \frac{E[R_p] - R_f}{\sigma_p} \\ \text{Fitness}(\vec{w}) &= 4.0 \cdot S_p + 2.5 \cdot E[R_p] - 2.0 \cdot \sigma_p \end{aligned}$$

5. Interactive Features in the App

- Step 1: Monthly Investment Slider:** Choose your SIP amount (₹2,000 to ₹1,00,000/month).
- Step 2: Goal Time Horizon:** Select 1 to 20 years with quick shortcuts (*3y Car, 5y Home, 10y Education, 15y Retirement*).
- Step 3: Risk Comfort:** Choose *Safe & Steady* 🟢, *Balanced* 🟡, or *Aggressive* 🔴.
- Step 4: Pick Your Leader:** Switch between α , β , and δ with 1 tap.
- Exact Rupee Breakdown:** Shows exactly how many rupees to put in Nifty 50, Mid Caps, Debt, Gold, and Global Tech every month.
- Future Wealth Growth Card:** Projects your exact terminal corpus value and total profit.

6. Real-World Example (₹15,000/month for 5 Years)

User Input:

- Monthly SIP:** ₹15,000 / month
- Horizon:** 5 Years (60 Months)
- Risk:** Balanced Growth

👑 Alpha Leader Output:

- Expected Return:** 13.8% / year
- Portfolio Volatility (Risk):** 12.4% (Moderate)

- Sharpe Ratio: 1.15

Monthly Asset Split:

- ● Large Cap (Nifty 50): 40% = ₹6,000/mo
- ● Mid & Small Cap Growth: 20% = ₹3,000/mo
- ● Corporate Bonds & Debt: 20% = ₹3,000/mo
- ● Sovereign Gold ETF: 10% = ₹1,500/mo
- ● Global / US Tech Index: 10% = ₹1,500/mo

💰 5-Year Wealth Result:

- Total Money Invested: ₹9,00,000
- Estimated Future Wealth: ₹12,85,400
- Total Profit Gained: +₹3,85,400 (+1.43x Multiple) 🎉

7. Why GWO Beats Traditional Portfolio Calculators

DIMENSION	TRADITIONAL FIXED RULES	BASIC SIP CALCULATOR	FINEXA ALPHAPACK™ (GWO)
Multi-Asset Intelligence	✗ None	✗ Single asset assumption	✓ 5-Asset Covariance Balancing
Optimization Method	✗ Guesswork	✗ Static compound formula	✓ Grey Wolf Pack Swarm Optimization
Leader Choices	✗ One-size-fits-all	✗ Single number	✓ 3 Leaders (\$\alpha\$, \$\beta\$, \$\delta\$) to fit personality
Downside Protection	✗ Ignored	✗ Ignored	✓ Bonds & Gold Volatility Dampener
User Friendliness	⚠ Confusing	⚠ Basic	✓ Clean cards, sliders & plain-English advice

Report Generated for FINEXA AI — Autonomous Wealth Systems.